



Lesson 5.3 — Cube Root Functions

Big idea: $f(x) = \sqrt[3]{x}$ is defined for all real numbers and has odd symmetry. Its graph passes smoothly through the origin and extends infinitely in both directions.

Quick review

Parent: $f(x) = \sqrt[3]{x}$. Domain $\mathbb{R}$, range $\mathbb{R}$. Passes through $(-1, -1)$, $(0, 0)$, $(1, 1)$.

Symmetry: $\sqrt[3]{-x} = -\sqrt[3]{x} \rightarrow$ odd function (180° rotational symmetry around origin).

Shape: increasing on all reals, with an inflection point at the origin.

Transformations: same form $g(x) = a\sqrt[3]{b(x - h)} + k$. Inflection point sits at (h, k) .

Worked example

Worked example. Sketch $f(x) = \sqrt[3]{x - 2} + 1$. State its inflection point and a couple of plotted points.

Inflection: $(2, 1)$. Points: $x = 1 \rightarrow \sqrt[3]{-1} + 1 = 0$. $x = 3 \rightarrow \sqrt[3]{1} + 1 = 2$. $x = 10 \rightarrow \sqrt[3]{8} + 1 = 3$.

Warm-ups

1. Compute $\sqrt[3]{8}$ and $\sqrt[3]{-8}$.
2. What is the domain of $f(x) = \sqrt[3]{x}$?
3. Where is the inflection point of $f(x) = \sqrt[3]{x + 4}$?

Practice

1. State the domain and range of $f(x) = 2\sqrt[3]{x - 1} + 3$.
2. Sketch (describe) $f(x) = -\sqrt[3]{x}$.
3. Find the inverse of $f(x) = \sqrt[3]{x + 2}$.

Challenge

1. If a cube has volume V (in^3), its side length is $s = \sqrt[3]{V}$ (in). Sketch (describe) s as a function of V on $V \geq 0$. Find s when $V = 64$. What V gives $s = 5$?
2. Show that $g(x) = \sqrt[3]{x}$ is the inverse of $f(x) = x^3$, and verify $f(g(x)) = x$ and $g(f(x)) = x$ for $x = 2$ and $x = -2$.



Answer key — Lesson 5.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. 2 and -2 .
2. All real numbers.
3. $(-4, 0)$.

Practice

1. Domain: $\mathbb{R}$. Range: $\mathbb{R}$.
2. Reflection of $\mathbb{R}x$ across the x -axis (so increasing becomes decreasing).
3. $y = \mathbb{R}(x + 2) \Rightarrow$ swap and cube: $x^3 = y + 2 \Rightarrow y = x^3 - 2$. $f^{-1}(x) = x^3 - 2$.

Challenge

1. Concave down, increasing; passes through $(0, 0)$. At $V = 64$: $s = 4$. For $s = 5$: $V = 125$.
2. $f(g(2)) = (\mathbb{R}2)^3 = 2$; $g(f(2)) = \mathbb{R}8 = 2$. Same with -2 : $\mathbb{R}(-2)^3 = -2$ and $(\mathbb{R}(-2))^3 = -2$. ✓